

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.5: Building Recommender Systems

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Research group

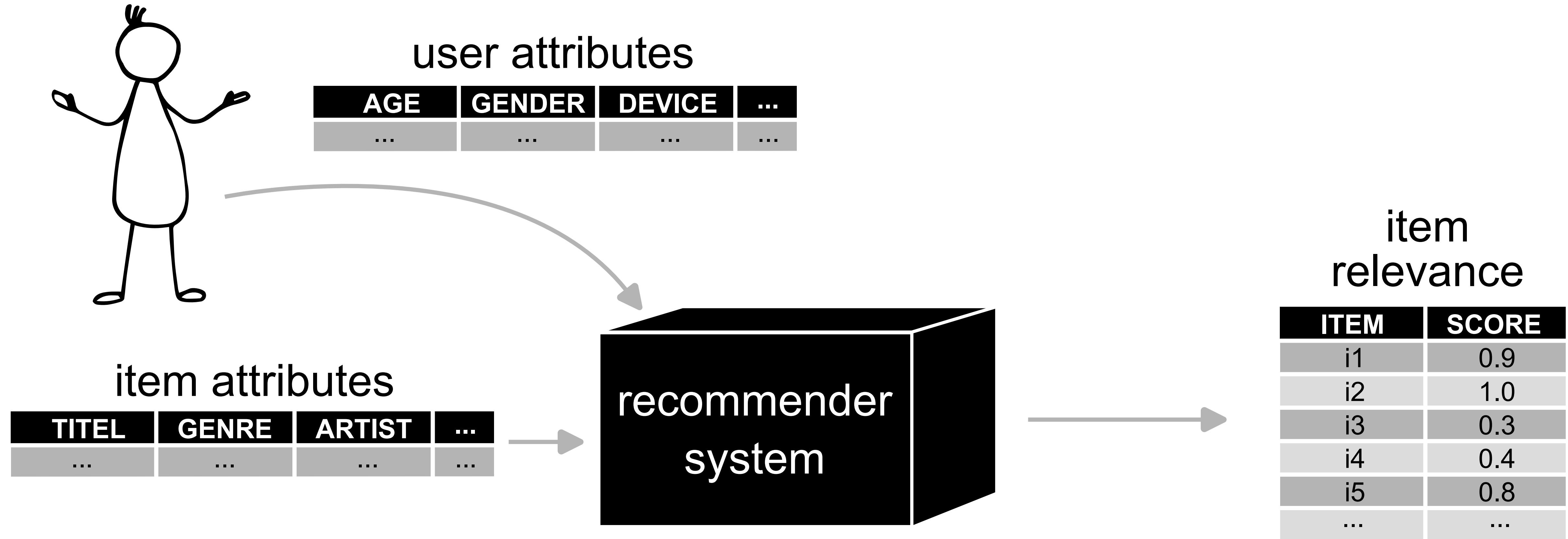
Complex Social & Computational Systems (CS)²

IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz

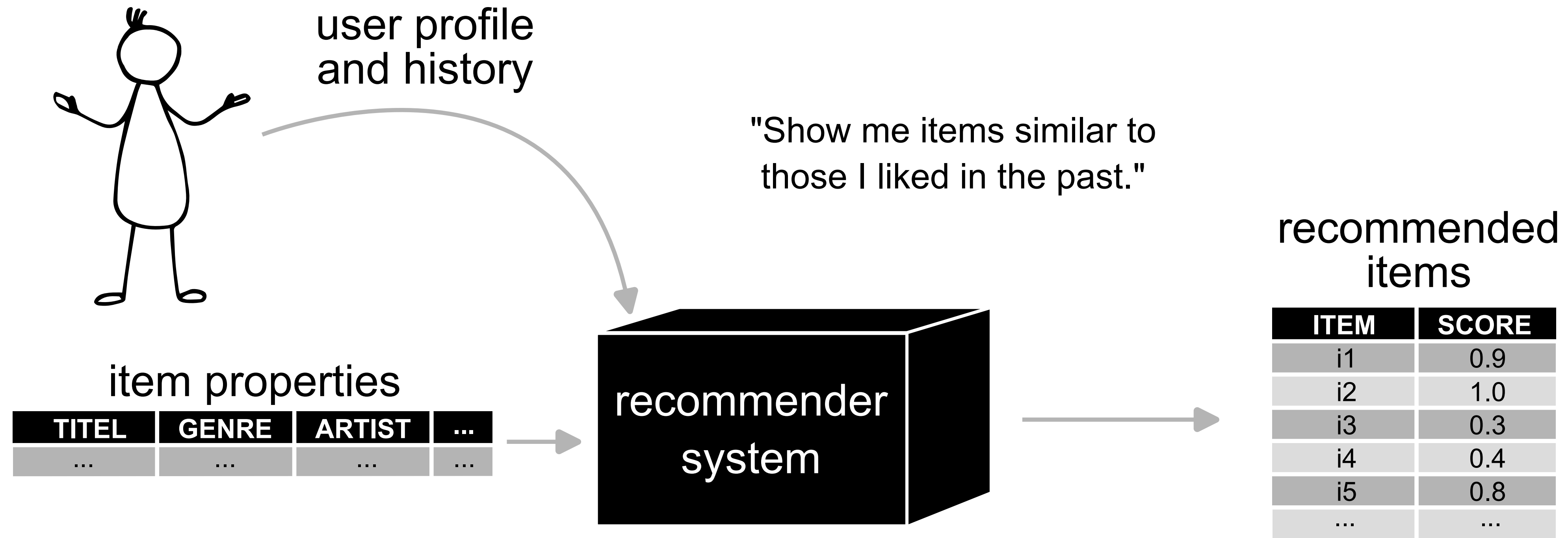


Basic Principle: Personalized Recommendation



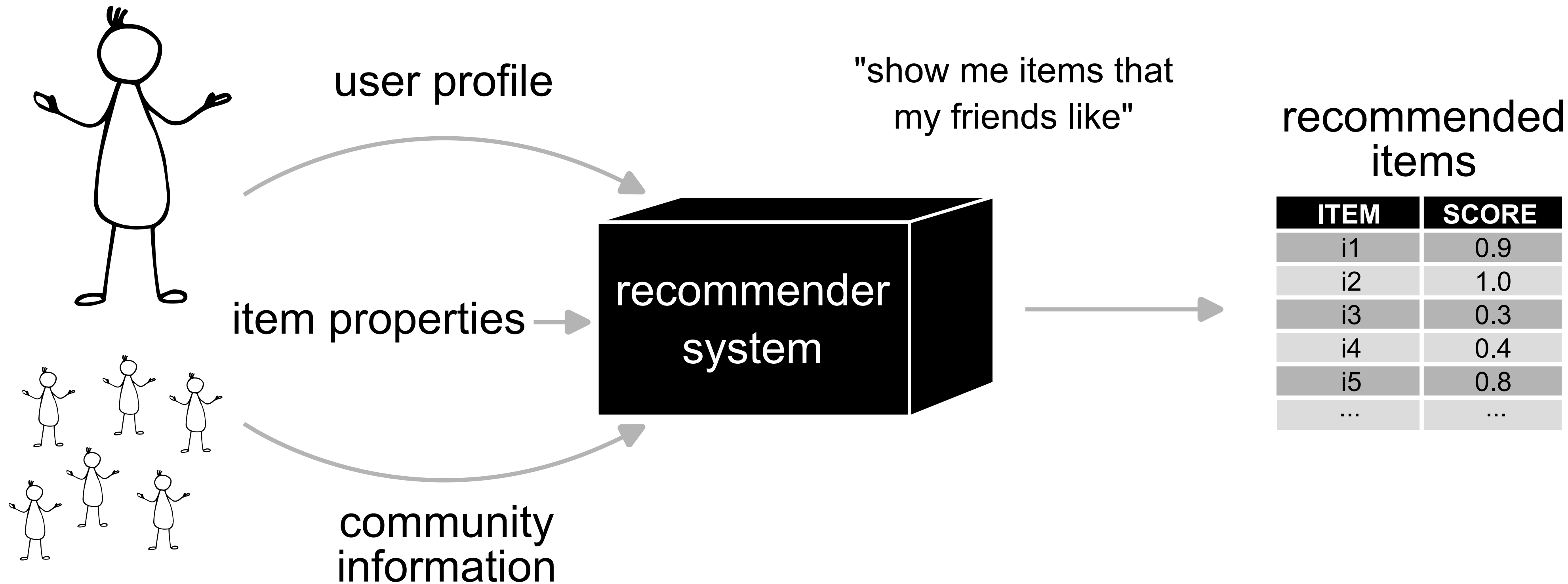
- Recommendation: Matching user profile and item profile.
- Single recommendation, no further interaction, no adaptation to behavior.

Content-Based Recommendation



- Automated construction of a user profile based on the **characteristics of the items** with which the user has interacted.
- Recommendation: Matching user history and item profile.
- Assumption: User preferences in the future are similar to preferences in the past.

Collaborative Filtering



- Using the opinion or behavior of **other users** for recommendation.
- Recommendation: matching similar users and recommending items other users liked.
- Assumption: Similar users like similar items.

Collaborative Filtering: Implementation Idea

We formulate the problem as "completing a table":

- We have information about how well our users rate different items (ratings from 1 to 5).
- For user Alice, we are missing a rating for item 5.
- Idea: we use the ratings of other users who are **similar** to Alice to **predict** Alice's rating for item 5 → supervised learning.

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

Collaborative Filtering: Algorithm

1. Find users who are similar to Alice, meaning those who liked the same items similarly well.
2. Use the average rating of these users for the missing item to predict whether Alice will like it too.
3. Follow this procedure for all items Alice has not yet seen and recommend the item with the best predicted rating.

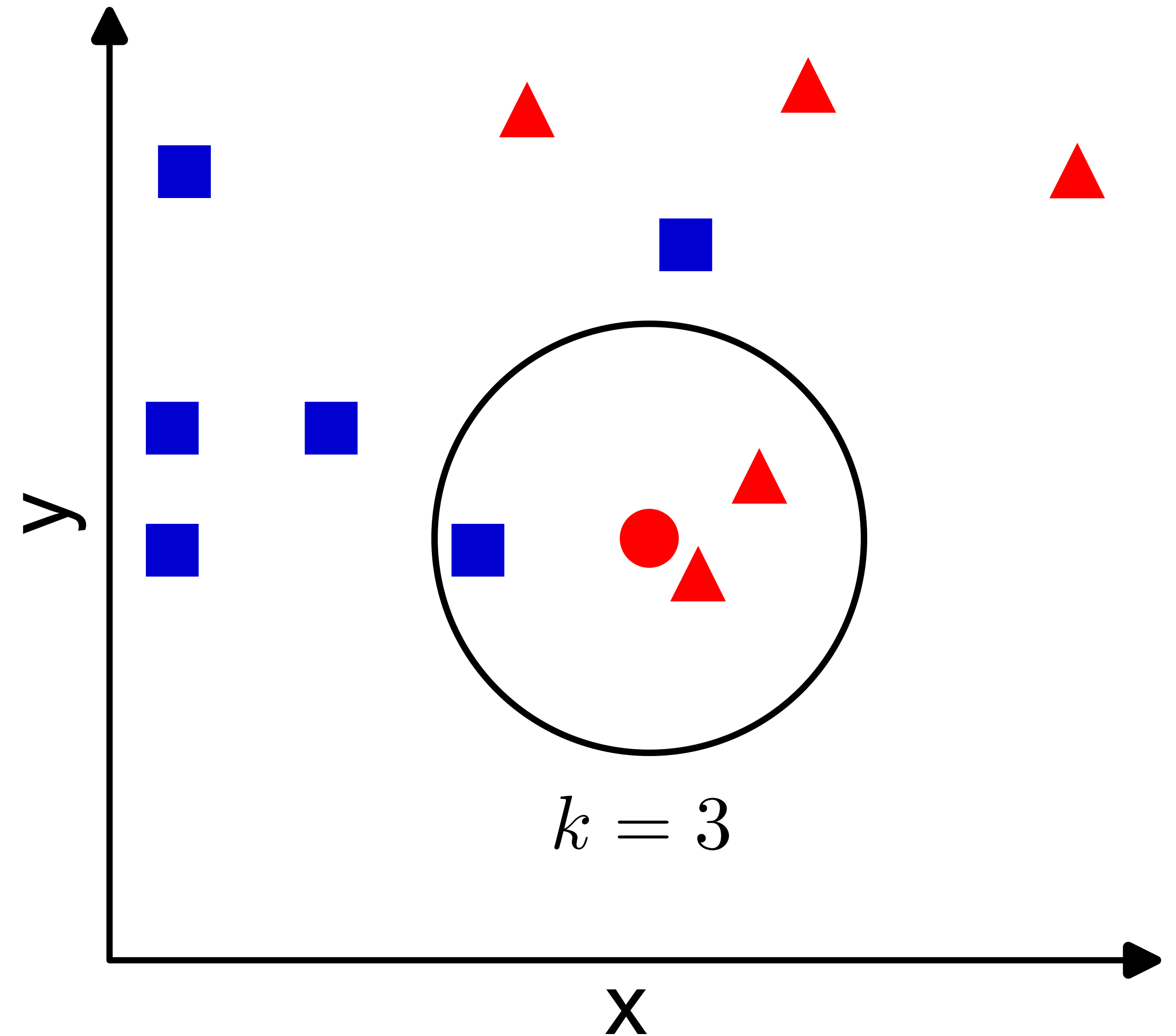
How well does the algorithm work?

→ Remove some of Alice's known ratings and predict them with this method.
Calculate the prediction error with the root mean squared error.

K-Nearest Neighbors Algorithm

Reminder: To classify a new observation:

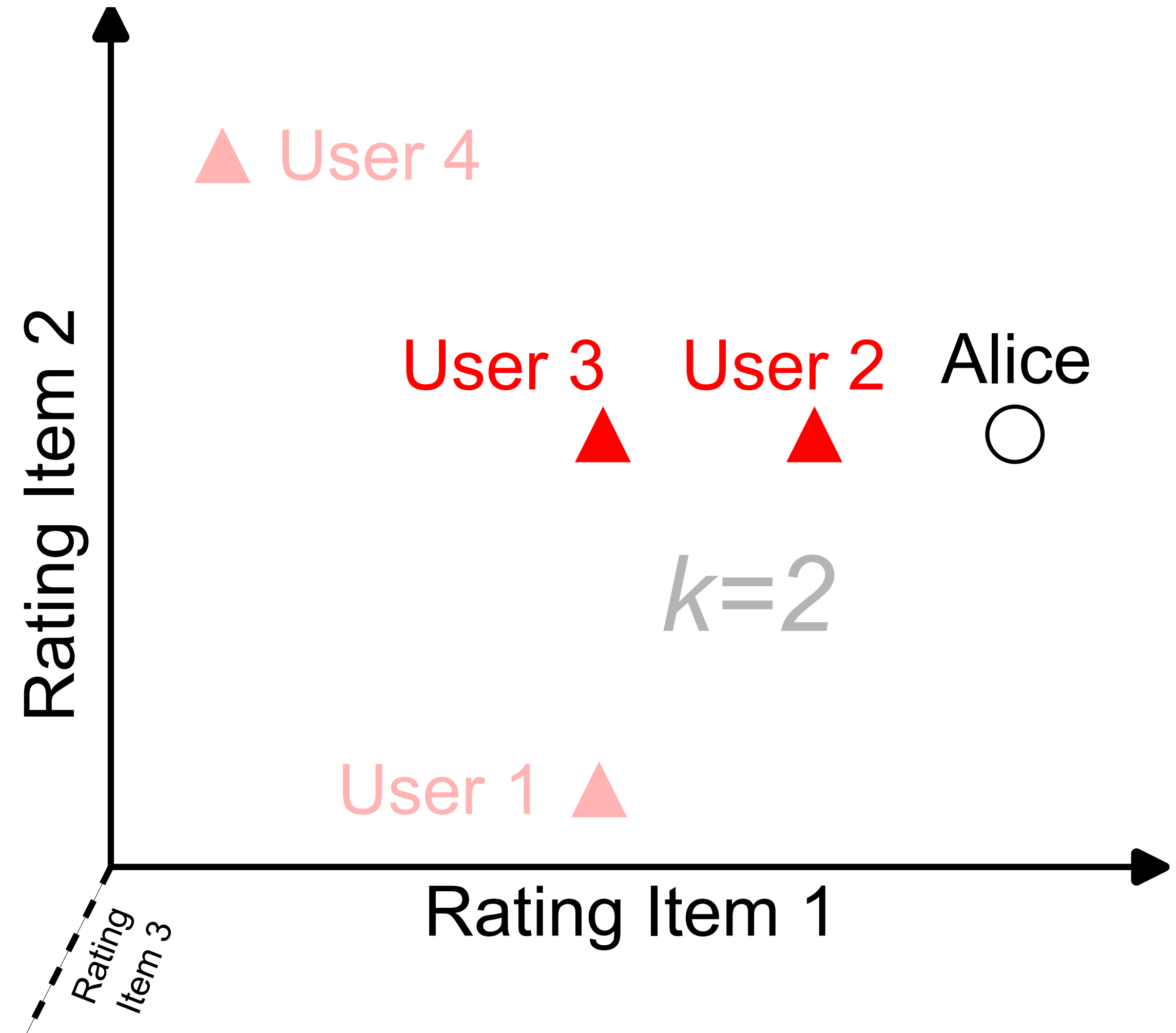
- Find the k nearest data points (" k -nearest neighbors").
- Choose the class that the majority of the nearest neighbors have (majority vote).



Identifying Similar Users

- Users are characterized by their ratings for items.
- Use e.g., **cosine similarity** to measure similarity between users.
- Find the k **users** most similar to Alice.
- Use the ratings of these users to predict ratings for items Alice has not yet seen.

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	4.5
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1



Summary

- Recommender systems can use information from **users, item properties, and information from the user community** for recommendations.
- Three approaches:
 - Personalized recommendation
 - Content-based recommendation
 - Collaborative filtering
- In most modern recommender systems, all three approaches are used simultaneously.